pairs of answers. Then, it finds the most similar answer, logs it, and outputs that as the answer.

Listing 10.18 `consistency.py`

```
from promptflow import tool
from typing import List
import numpy as np
from scipy.spatial.distance import cosine
@tool
def consistency(texts: List[str],
                embeddings: List[List[float]]) -> str:
    if len(embeddings) != len(texts):
        raise ValueError("The number of embeddings ↪
       ↪ must match the number of texts.")

    mean_embedding = np.mean(embeddings, axis=0)     #1
    similarities = [1 - cosine(embedding, mean_embedding) ↪
                ↪ for embedding in embeddings]     #2
    most_similar_index = np.argmax(similarities)     #3

    from promptflow import log_metric
    log_metric(key="highest_ranked_output", value=texts[most_similar_index])     #4

    return texts[most_similar_index]     #5
```

#1 Calculates the mean of all the embeddings
#2 Calculates cosine similarity for each pair of embeddings
#3 Finds the index of the most similar answer
#4 Logs the output as a metric
#5 Returns the text for the most similar answer

We need to run the evaluation flow in batch mode as well. Open `self-consistency-evaluation/flow.dag.yaml` in VS Code and run the flow in batch mode (beaker icon). Then, select Existing Run as the flow input, and when prompted, choose the top or the last run you just executed as input.

Again, after the flow completes processing, you'll see an output like that shown in listing 10.17. Ctrl-click on the output folder link to open a new instance of VS Code showing the results. Locate and open the `metric.json` file in VS Code, as shown in figure 10.9.